

The Register, Out Loud

Runs ~12 min + questions Open alongside the artifact If time is short cut beats 3 and 5 New since it was written beat 6

Open with this

“A mentor told me I was using papers to justify what I already believed. He was right, and the fix changed how I run every experiment since: write down what would prove you wrong, before you run anything.”

Then open the artifact and scroll it as you go. Each beat below is one of its sections.

The beats

0:00–1:00

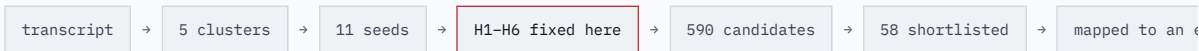
Where this came from

- I presented the literature review to the cohort: 590 candidate papers narrowed to 58, six hypotheses, five clusters.
- Five things were said. One landed on the method itself, and it was the useful one.
- This artifact is my written answer: what I accepted, what I pushed back on, and what changed as a result.

1:00–3:00

The criticism that landed

- “I feel that you are trying to justify your hypothesis with papers.”
- Look at the order I worked in: transcript, then five clusters, then eleven seed papers, then *six hypotheses fixed*, then 5 candidates, then 58 shortlisted, then each one mapped to a hypothesis I already had.
- My spreadsheet had three judgement columns. All three asked: how does this paper relate to something I already dec
- **There was no column where a paper could say “here is something you haven’t thought of.”** So that answer could ne no matter what the papers said.



Say the honest limit too: it does not invalidate the corpus. Three of the review’s main findings landed against what I exp is not what a self-confirming review produces.

3:00–5:00

The other three things, and where I pushed back

- **“How can you say H3 and H5 are unclaimed?”** Fair as stated. My evidence was thin coverage — six and seven papers — which is not absence. And my search method is known to under-sample this benchmark: 512 backward references and 8 forward lists surfaced *none* of the 13 ARC-AGI-3 papers, because no seed cites the benchmark. I downgraded the wording to “not found in this corpus”.
- **“H5 and H6 are a single hypothesis.”** Half right. They are separable mechanisms — one prunes before acting, one fires after a reset — but as an experiment they’re one harness with two switches. Running them apart doubles the compute for a distinction the metric may not resolve.

- **“Look outside machine learning.”** The sharpest addition. My five clusters came from watching humans play the game, and I then answered them entirely with ML papers. Developmental psychology, Gibson on affordances, Ashby on requisite variety, Powers on perceptual control — all unread.

Worth saying: Gibson is a live threat to H3, which assumes controllability is something you go and identify. Better to know that before building it.

5:00–7:30

What replaced the ranking: the register

- I had been planning to write “which hypothesis is most promising”. That's an opinion dressed as a deliverable.
- Instead: every experiment is written down before it runs, with the result that would prove it wrong.
- A block names the baseline, the treatment, the primary metric, the falsifier, and the abandon condition — a compute budget after which I stop regardless.
- It also fixes what could otherwise be fudged later: which games are held back, how many seeds, what counts as a failed run.
- And it reports a *vector*, not a score: an idea that helps one cluster may hurt another, and a single number hides that.

The rule that matters: a block can be edited freely until its first run. After that, changes are appended as dated amendments, never edited in place.

7:30–9:00

The counter-intuitive bit: a small model is the method

- People assume you'd want the biggest model. The literature says otherwise for *measuring*.
- One paper ablated the reference architecture piece by piece, scoring 34 to 75. The spread from changing the model was larger than the spread from changing the architecture.
- So on a frontier model, your harness's effect is buried under the model's own capability. You can't tell a real gain from noise.
- Freeze a small model and the harness becomes the biggest thing in the measurement. The competition's no-internet rule doesn't force a worse experiment — it forces the only one that can answer the question.

State the risk up front: the harness may help only small models. That's a real result, and I wrote it down before running anything.

9:00–11:00

How it has aged — the part that isn't in the artifact

- Written on 10 September. Since then it has been used three times, and it did its job every time.
- **It caught a dead metric.** EO's falsifier said the experiment is meaningless if the official score reads zero everywhere. A random agent scores zero on 23 of 25 games, so the finer-grained measures were promoted *before* any model ran.
- **It stopped me keeping the good half.** This week's fix was written as a test first: the click game must improve, the keyboard game must not get worse. Click improved, 52.5 to 70.6 against random's 77.5. The control came back inconclusive — and it's recorded as inconclusive, not as a win.
- **It made two wrong guesses cheap.** I predicted the model was repeating one move, then that it was oscillating. Both were tested on my laptop with no GPU, and both were refuted before spending anything.
- **And the wording change was vindicated.** The June milestone write-up shows second place using something close to H5. “Not found in my corpus” was right; “unclaimed” would have been wrong.

11:00–12:00

Close on the habit, not the project

- The transferable thing isn't the ARC work. It's one line: write down what would prove you wrong, before you run anything.
- It costs a few minutes. It converts an opinion into something that can fail.

- And it makes a negative result publishable instead of embarrassing — which is why I can stand here and tell you the model lost to random.

Numbers, in one place

590 → 58 Candidates narrowed to the shortlist	0 of 13 ARC-AGI-3 papers the citation graph found; all came from direct search	6 and 7 Papers supporting H3 and H5 — thin, not absent	34–75 Ablation scores where model choice outweighed architecture
13 of 56 Papers that ran open-weight models, none on this benchmark	0.04 Levels a random agent clears per game — the bar	0 / 16 Games where the first real model beat random	52.5 → 70.6 Click-game exploration after this week's fix (random: 77.5)

Questions to expect

Doesn't this just slow you down?

Writing a falsifier takes minutes. It has already saved two GPU runs and stopped me reporting a half-result as a win. The slow path is discovering after the fact that your measurement couldn't have shown anything.

If the review was biased, why keep the 58 papers?

The defect was narrow: no channel for a paper to propose something new. The papers were selected against the clusters, which came from humans playing the game, not from my hypotheses. And several findings landed against my expectations. The fix is one backward pass over data I already have.

Isn't "not found in my corpus" just hedging?

It's the difference between two claims, and one of them turned out false. Second place in the June milestone used something close to the hypothesis I'd called unclaimed. Precise wording is why I don't have to retract anything.

Why not just use the biggest model and win?

Two different goals. To score, yes, use 27–31B like the winners. To learn whether *my idea* helps, a big model hides the effect. I'm running both sizes for exactly that reason.

What if all your experiments fail?

Then I publish that, with a clean baseline and pre-registered falsifiers. The register has a stop condition saying so. A negative result with an honest method is a milestone deliverable, not a failure.

Can I use this?

Yes, and it needs no tooling. One file, one block per experiment: baseline, treatment, primary metric, falsifier, abandon condition. Dated. Committed before the run.

If you're asked something you don't know

Say so and point at where it's written. Every decision and result is recorded with its source, so “I don't remember, it's in the register” is a real answer.

Don't oversell the register. It doesn't make the work correct; it makes the work checkable. The model still lost to random this week.

Two things genuinely open: the cross-domain reading — psychology, cybernetics — hasn't been done yet; and the backward pass over the 58 papers is written down but not run.

Companion to *What Would Prove It Wrong* (artifact 920a5e75), itself the response to the week-15 peer session. Sources:

`research_prep/notes/peer_review_mapping.md` , `research_prep/notes/experiment_register.md` . Beat 6 reflects runs completed after the artifact was written.

Compiled 12 September 2026.